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Research Article

Keywords:

Posted Date: May 10th, 2023

DOI: https://doi.org/10.21203/rs.3.rs-2878411/v1

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Additional Declarations: No competing interests reported.

Version of Record: A version of this preprint was published at Brain Topography on September 26th, 2023. See the published version at https://doi.org/10.1007/s10548-023-01006-2.

Complexity measures for EEG microstate sequences - concepts and algorithms

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Abstract

EEG microstate sequence analysis quantifies properties of ongoing brain electrical activity which is known to exhibit complex dynamics across many time scales. In this report we review recent developments in quantifying microstate sequence complexity, we classify these approaches with regard to different complexity concepts, and we evaluate excess entropy as a yet unexplored quantity in microstate research.

We determined the quantities entropy rate, excess entropy, Lempel-Ziv complexity (LZC), and Hurst exponents on Potts model data, a discrete statistical mechanics model with a temperature-controlled phase transition. We then applied the same techniques to EEG microstate sequences from wakefulness and non-REM sleep stages and used first-order Markov surrogate data to determine which time scales contributed to the different complexity measures.

We demonstrate that entropy rate and LZC measure the Kolmogorov complexity (randomness) of microstate sequences, whereas excess entropy and Hurst exponents describe statistical complexity which attains its maximum at intermediate levels of randomness. We confirmed the equivalence of entropy rate and LZC when the LZ-76 algorithm is used, a result previously reported for neural spike train analysis (Amigó et al, 2004). Surrogate data analyses prove that entropy-based quantities and LZC focus on short-range temporal correlations, whereas

Hurst exponents include short and long time scales. Sleep data analysis reveals that deeper sleep stages are accompanied by a decrease in Kolmogorov complexity and an increase in statistical complexity.
Regarding the practical use of these methods, we suggest that LZC can be used as an efficient entropy rate estimator that avoids the estimation of joint entropies, whereas entropy rate estimation via joint entropies has the advantage of providing excess entropy as the second parameter of the same linear fit.
We conclude that metrics of statistical complexity are a useful addition to microstate analysis and address a complexity concept that is not yet covered by existing microstate algorithms while being actively explored in other areas of brain research.

Introduction

EEG microstate analysis has become a widely used method to characterize spontaneous and evoked brain activity patterns (Michel and Koenig, 2018). Complexity is a hallmark of integrated, large-scale brain activity and researchers have an intuition of how complexity manifests in their data, may that be neuronal spiking (Amigó et al, 2004; Szczepański et al, 2004), local field potentials (Abásolo et al, 2014), functional neuroimaging (Xin et al, 2021; Nezafati et al, 2020; Hancock et al, 2022), electroencephalography (EEG) (Casali et al, 2013), or magnetencephalography (MEG) (Fernández et al, 2011).

Yet, there is no unique theoretical notion of what complexity is or how it should be measured, and the fact that complexity is a multifaceted concept is reflected in the existence of a large number of complexity definitions in the literature (Shalizi, 2006). Definitions of complexity have emerged in different scientific disciplines, and some concepts have re-emerged under different names which makes it challenging to maintain an overview of this research area (Prokopenko et al, 2008; Ay et al, 2011; Crutchfield and Feldman, 2003). We will review the complexity concepts that have so far been used to characterize EEG microstate sequences, contextualize these concepts in the wider field of complexity research, and explore links between these measures and other areas of brain research. We test excess entropy, a measure which has not been applied to microstate sequence analysis yet, and demonstrate that it fills a gap in the information-theoretic description of microstate sequences.

Concepts of complexity

There are at least two basic flavours of complexity, each founded on different intuitions about what represents complexity.

One concept is known as algorithmic or Kolmogorov complexity (Alekseev and Yakobson, 1981) and initially defined complexity as the length of the shortest program that is able to reproduce the input data. This measure increases monotonically with the amount of 'randomness' in the data. Intuitively, the shortest algorithm reproducing a random sequence is a command that prints exactly that sequence and the length of that algorithm would be approximately the same as the length of the dataset. In

the microstate context, this would correspond to a sequence of independent samples from the set of microstate labels, e.g. from $\{A, B, C, D\}$, possibly weighted by their relative occurrence. At the non-random end of the spectrum, a sequence that consists of a single repeated label, e.g. $AAA\ldots$, is reproduced by a very short instruction such as 'print A, n times'. As will be explained further below, there are more practical approaches to measure Kolmogorov complexity than trying to find the actual program, namely entropy rate and Lempel-Ziv complexity.

A separate family of complexity measures was developed as many researchers are uncomfortable with the concept of complexity being essentially the same as randomness. Following the Kolmogorov complexity concept, randomly connected neurons are more complex than a real brain, and electrical white noise is evaluated as being more complex than actual brain electrical activity. The 'statistical complexity' concept asks how difficult it is to obtain a statistical model of the data and aims to construct bell-shaped complexity measures when plotted against randomness (Huberman and Hogg, 1986; Lindgren and Nordahl, 1988). Both complexity concepts agree in that extremely ordered systems should be assigned a low complexity, as they are non-random and a compact model can often be formulated. An example of a highly ordered spatial system is a crystal structure, as is a sine wave in the world of time series. Where the two complexity concepts differ is in their assessment of highly random systems. They have high Kolmogorov complexity but their statistical complexity is low because a simple assumption (statistical independence) can already provide a good statistical model of the data (Grassberger, 1986; Crutchfield, 1994). A similar line of reasoning led to the concept of Tononi-Sporns-Edelman complexity which has attained attention as a theory of consciousness and brain function in general (Tononi et al, 1994).

In Fig. 1 these concepts are illustrated with snapshots of the Potts model that will be formalized further below. The model attains one of five discrete states on each node of a regular square lattice and the dynamics are controlled by a parameter that corresponds to thermodynamic temperature. From left to right, with increasing temperature, the spatial patterns become increasingly random and disordered. This reflects Kolmogorov complexity, which increases monotonically with temperature. Statistical complexity, however, has its peak around the critical temperature of the model ($T = T_c$) where a phase transition occurs. Around T_c, the most extensive spatial and temporal correlations occur and statistical forecasting of the model states is most challenging.

The two complexity concepts discussed can be quantified by a range of metrics that require some disambiguation of the historical terminology. Kolmogorov complexity can be estimated as the randomness of a signal after all correlations have been taken into account (irreducible randomness (Prokopenko et al, 2008)). This is captured by the entropy rate, which is derived from joint entropy (or block entropy) estimates of signal subsequences of different lengths, or via Lempel-Ziv complexity which measures this type of complexity by compressing the signal on the basis of repeated patterns (Lempel and Ziv, 1976). The more a signal can be compressed, the lower its algorithmic (Kolmogorov) complexity. Entropy rate is also known as Kolmogorov-Sinai entropy in the field of dynamical systems and chaos theory (Shalizi, 2006).

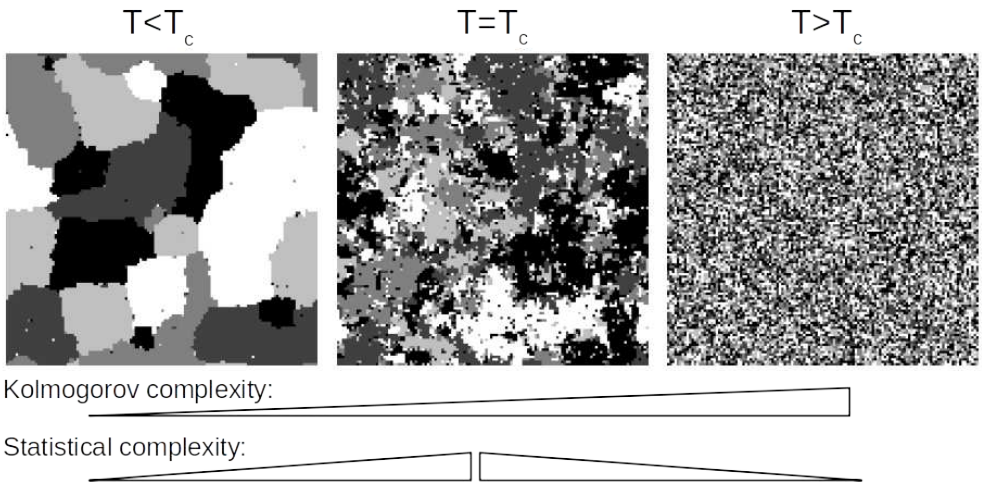


Fig. 1 Two complexity concepts illustrated with snapshots of a 2D Potts lattice model ($Q = 5$, 128×128 nodes) at different temperatures T. **Left**: The Potts model ($Q = 5$) at low temperature ($0.6 \times T_c$, T_c is the critical temperature) shows a clear ordering related to low Kolmogorov complexity and low statistical complexity. **Center**: Close to the phase transition ($T = T_c$), where the most extensive spatial and temporal correlations occur, we find intermediate Kolmogorov complexity and maximum statistical complexity. **Right**: Above the critical temperature ($3.0 \times T_c$) spatial features are apparently random (low order) but a simple independence assumption provides a good statistical model for the system, a situation characterized by large Kolmogorov complexity (randomness) and low statistical complexity.

Statistical complexity can be assessed by different measures. One of these measures was called statistical complexity in (Crutchfield and Young, 1989), but had already appeared as true measure complexity in (Grassberger, 1986). This quantity is based on a graph model of the underlying process and often needs to be reconstructed from data. As model reconstruction is not trivial, approximations can be studied instead (Grassberger, 1986; Prokopenko et al, 2008). A lower bound to the true measure complexity was named effective measure entropy in (Grassberger, 1986) and excess entropy in (Crutchfield and Feldman, 1997). We will use the latter term as it is used in the recent literature and expresses Grassberger's initial observation that finite estimates of the entropy rate converge slowly towards the asymptotic value. Excess entropy can be expressed as the surplus of entropy across all finite entropy rate estimates (Grassberger, 1986; Crutchfield and Feldman, 2003). The same quantity appeared as predictive information in (Bialek et al, 2001) and the name refers to the interpretation that statistical complexity measures the predictability of a time series. This idea is also expressed by its information-theoretic definition as the shared information between the past and the future of a signal, relative to an arbitrary observation time point. The concept of statistical complexity has therefore also been named forecasting complexity (Zambella and Grassberger, 1988).

Complexity and microstate research

The methods that have been used in both, EEG microstate research and the wider field of complexity research, are entropy rate, Lempel-Ziv complexity (LZC), and Hurst exponents. Hurst exponents were first used for microstate analysis by Van De Ville et al. (Van de Ville et al, 2010). Although the aim of the authors was to address self-similarity and fractality rather than explicitly measuring complexity, the relationship between Hurst exponents and complex systems properties is present throughout the article. We evaluated their approach in relation to Markov models of microstate sequences in (von Wegner et al, 2016) and applied it to cognitive load assessment in (Jia et al, 2021).

Entropy rate estimation for microstate sequence analysis was introduced in (von Wegner et al, 2018a), and we evaluated its changes during different types of cognitive effort in (Jia et al, 2021), and for NREM sleep stages in (Wiemers et al, 2022). We interpreted entropy rate in terms of sequence predictability in (von Wegner et al, 2018a) and (Jia et al, 2021), and as a complexity measure in (Wiemers et al, 2022).

Next, Lempel-Ziv complexity analysis of microstate sequences was introduced in (Tait et al, 2020), where a loss of complexity in the EEG of Alzheimer disease patients was demonstrated by applying a quantity called Omega complexity to the raw EEG signal (Wackermann, 1996) and LZC to microstate sequences. A more recent variant of the Lempel-Ziv algorithm was used subsequently in (Artoni et al, 2022) where concentration-dependent effects of propofol on microstate LZC were investigated. An intermediate approach can be found in (Irisawa et al, 2006), where the EEG signal was classified with Omega complexity, and the results were compared to microstate duration but complexity analysis was not applied to microstate sequences.

Outline

Evaluating complexity metrics on neurophysiological data alone leaves some uncertainty about what is actually measured, as the full brain state is not well defined given a lack of reference measurement. For this reason, we first evaluate the selected metrics (entropy rate, excess entropy, Lempel-Ziv Complexity, and Hurst exponents) on a well-understood numerical model from statistical physics, the discrete Potts model (Wu, 1982). The Potts model offers some advantages in this context. First, the model produces time series over a discrete state space with an arbitrary number of states, similar to EEG microstate sequences. The number of states is often denoted Q for the Potts model, and K in microstate research, related to the initial K-means clustering of EEG data. The second advantage of the Potts model is that there is a single control parameter (temperature) that controls the appearance of a phase transition. The common elements between the Potts model and EEG microstates are a) entropy, which is closely linked to temperature in statistical physics, but also has an interpretation in terms of time series predictability, and b) phase transitions have been discussed as an important feature of resting-state brain activity and are often quantified by complexity metrics such the Hurst exponent, for example in EEG research (Linkenkaer-Hansen et al, 2001; Kantelhardt et al, 2015; von Wegner et al, 2018b),

EEG microstate research (Van de Ville et al, 2010; Jia et al, 2021), and fMRI studies (Bullmore et al, 2009; Tagliazucchi et al, 2013).

In the second part of the results section, we evaluate the four metrics on EEG microstate sequences in wakefulness and non-REM (NREM) sleep, a dataset we have previously analyzed with other microstate analysis tools (Brodbeck et al, 2012; Wiemers et al, 2022). In an attempt to identify which time series features the different complexity metrics actually 'see', we use first-order Markov surrogate data to represent exactly that amount of information that is captured by the transition probability matrix, an approach that is often used to report microstate data (Lehmann et al, 2005).

Methods

Computational model

In microstate research, K-means clustering is commonly performed for $K = 4$ or $K = 5$ clusters. Hence, we implemented the Potts model with Q states for $Q = 4, 5$ as reviewed in (Wu, 1982) on a two-dimensional (2D) discrete lattice geometry. Other topologies could be employed but the 2D model is well studied and the critical temperatures are known analytically (Wu, 1982; Brown et al, 2022). Two different energy (Hamilton) functions have been presented for the Potts model, the standard model and the vector (clock) model. We chose the standard model for which the type of phase transition is known (Wu, 1982).

The Potts model uses discrete variables that can be visualized as 2D unit vectors (spins), uniformly distributed around the complex unit circle. Their energy difference is defined by the phase difference. Formally, the spin values are given by $S = \{\exp(2\pi i q/Q),\, q = 0, \ldots, Q-1\}$ with phase $2\pi q/Q$. For our purpose, it is sufficient to store the integer values $q = 0, \ldots, Q-1$ as the discrete model states. In the standard Potts model, a lattice site (k, l) with spin phase ϕ_{kl} has energy:

$$E_{kl} = -\sum_{m,n \in N_{kl}} J\delta(\phi_{kl}, \phi_{mn}) \tag{1}$$

using the Kronecker delta function (δ) and nearest-neighbour coupling, i.e. the neighbours of spin ϕ_{kl} at lattice site (k, l) are $N_{kl} = \{(k-1, l), (k+1, l), (k, l-1), (k, l+1)\}$. We exclusively considered ferromagnetic coupling ($J = +1$) which favours neighbouring spins to align, as the lowest energies are produced when their phase values are identical. In a neuronal context, this can be interpreted as neuronal ensembles which tend to align the phase of their voltage oscillations (Breakspear et al, 2010).

We simulated the Potts model on a square lattice of 25×25 nodes. Model data were generated by Monte-Carlo simulation with Metropolis sampling, i.e. a randomly chosen $q \to q'$ transition was accepted with probability $p = \min(1, \exp\left(-\frac{\Delta E}{T}\right))$ which depended on the energy difference ΔE before and after the proposed transition. In words, transitions that reduced the lattice energy were always accepted whereas transitions increasing the total system energy were only accepted if they could jump across

the energy barrier ΔE, which was tested stochastically by comparison with a uniformly distributed pseudo-random number. The critical temperature of the two-dimensional Potts model is $T_c = \left(\log(1+\sqrt{Q})\right)^{-1}$ (Brown et al, 2022).

Individual simulations were run for $t = 30000$ iterations, preceded by a warm-up of 2500 iterations to allow relaxation from the initial random state. We simulated the system across a range of temperatures which will be written relative to the critical temperature T_c. We used relative temperatures T/T_c of 0.2, 0.4, 0.6, 0.8, 0.9, 1.0, 1.1, 1.2, 1.4, 1.6, 1.8, 2.0, 2.2, 2.4, 2.6, 2.8, 3.0. For each temperature, we ran the model 50 times and selected a subset of $n = 25$ random lattice nodes for complexity analysis. We chose a length of 30000 samples to match the length of our EEG segments (2 minutes of EEG acquired at 250 Hz).

Experimental data and EEG pre-processing

We analyzed EEG recordings from n=19 healthy subjects. The EEG dataset is a subset of the dataset analyzed in (Brodbeck et al, 2012; Wiemers et al, 2022) and only includes those subjects for whom sleep stage N3 data was available. Briefly, EEG data from simultaneous EEG-fMRI recordings were corrected for scanner and cardioballistic artefacts and downsampled to 250 Hz as described in (Brodbeck et al, 2012). Sleep stages were scored manually, according to international standards. The data were band-pass filtered to 1-30 Hz with a digital Butterworth filter of order six.

Microstate algorithm

Subject-wise microstates were computed with the modified K-means algorithm (Pascual-Marqui et al, 1995) implemented in Python 3 (von Wegner and Laufs, 2018). Group-wise microstate maps were computed for each sleep stage with a full permutation procedure over the subject-wise maps (Koenig et al, 1999), repeated 20 times with random initial conditions. Group maps were defined as the result that maximized the global explained variance across the subjects. Microstate sequences were obtained by competitive back-fitting at each time step and ignoring map polarity. No further smoothing methods were applied. Microstate maps and further properties of this dataset are detailed in (Wiemers et al, 2022).

Surrogate data

Surrogate data for the Potts model and EEG microstate sequences were synthesized as first-order Markov processes, based on the empirical transition probability matrix of each time series as explained in (von Wegner et al, 2016; von Wegner and Laufs, 2018).

Complexity metrics

Entropy rate and excess entropy

Entropy rate and excess entropy were computed as described in (von Wegner, 2018) and published in our 2017 Python microstate package (github repository) (von Wegner

and Laufs, 2018). Briefly, the frequency distribution of microstate sequence blocks ('microstate words') $\mathbf{X}_n^{(k)} = (X_n, \ldots, X_{n+k-1})$ for each block length $k = 1, \ldots, 6$ was estimated from the data, and the joint entropy $H\left(\mathbf{X}^{(k)}\right)$ of each distribution $P\left(\mathbf{X}^{(k)}\right)$ was computed . The parameters entropy rate (h_X) and excess entropy ($\mathbf{E}$) were obtained as the slope and y-axis intercept of a linear fit of $H\left(\mathbf{X}^{(k)}\right)$ vs. k, respectively. This approach is visualized in Figure 2 for three different situations of the Potts model. This estimate of the entropy rate h_X is based on the following definition in terms of infinitely long observations

$$h_X = \lim_{n\to\infty} \frac{1}{n} H(\mathbf{X}_n^{(k)}) \tag{2}$$

which, for stationary stochastic processes, is equivalent to the conditional entropy form

$$h'_X = \lim_{n\to\infty} H(X_{n+1}|\mathbf{X}_n^{(k)}). \tag{3}$$

The second form (3) can be read in terms of time series predictability; h_X expresses the uncertainty (entropy) in predicting the next state of the sequence (X_{n+1}) when the last k states ($\mathbf{X}_n^{(k)}$) are known.

In a similar approach, excess entropy can be expressed as the mutual information between the past $\mathbf{X}_{\text{past}} = (\ldots, X_{n-1}, X_n)$ and the future $\mathbf{X}_{\text{future}} = (X_{n+1}, X_{n+2}, \ldots)$ of the process

$$\mathbf{E} = I(\mathbf{X}_{\text{future}}; \mathbf{X}_{\text{past}}) \tag{4}$$

The concept can be made more intuitive by re-writing mutual information between the random variables X, Y as $I(X;Y) = H(X) - H(X|Y)$, i.e. the reduction in uncertainty about X by knowing Y. In this sense, excess entropy encodes to what extent predictions about the future improve, or entropy decreases, by including knowledge about the past of the process:

$$\mathbf{E} = H(\mathbf{X}_{\text{future}}) - H(\mathbf{X}_{\text{future}}|\mathbf{X}_{\text{past}}) \tag{5}$$

Lempel-Ziv complexity (LZC)

The 1976 implementation of the Lempel-Ziv algorithm (LZ-76, (Lempel and Ziv, 1976)) was written and compiled in Cython 0.29.3 to produce faster Python 3.6.9 code. Our implementation is identical to the algorithm used in (Tait et al, 2020) which is publicly available.

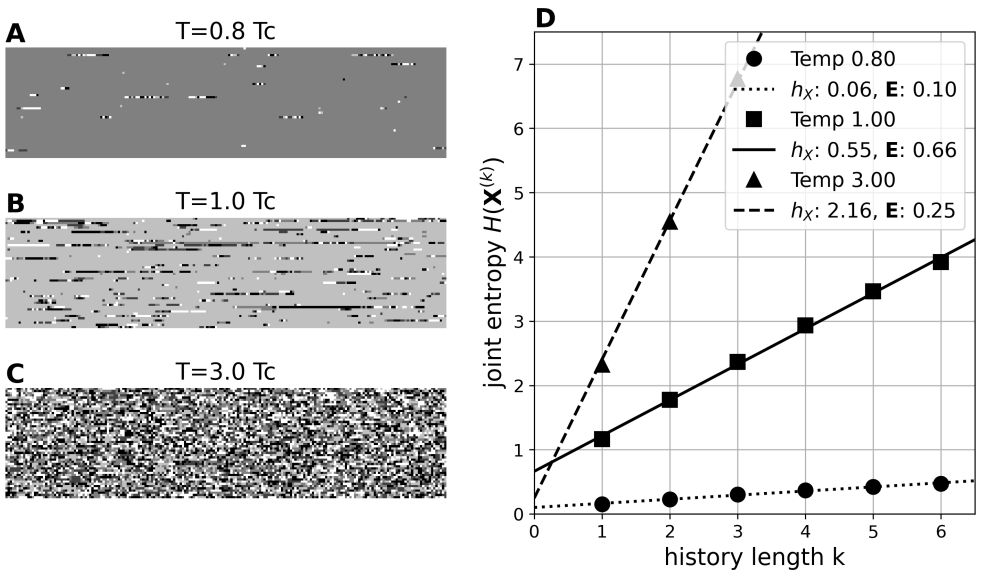


Fig. 2 Entropy rate and excess entropy estimation. On the left (A-C) time series at random lattice nodes of the Potts model ($Q = 5$) are illustrated as raster images, grey values correspond to the five Potts model states. Time runs from left to right, starting at the top row. **A**: Time series at a sub-critical temperature (0.8 T_c) have a relatively uniform appearance with long stretches of constant values and low randomness. **B**: At criticality ($T \sim T_c$), the Potts model time series show an intermediate level of randomness and maximum statistical complexity. **C**: At high temperatures (3.0 T_c) states appear in a random, unpredictable order but statistical time series properties can be modelled with a simple independence assumption, resulting in low statistical complexity. **D**: The properties described in A-C are captured by the slope (entropy rate h_X, randomness) and y-axis intercept (excess entropy **E**, statistical complexity) of the linear fit to the joint entropies $H\left(\mathbf{X}^{(k)}\right)$.

Hurst exponents

Hurst exponents were calculated by detrended fluctuation analysis (DFA) as described in (Van de Ville et al, 2010; von Wegner et al, 2016), using 50 logarithmically spaced time scales over the range of 50-2500 samples (200 ms - 10 s for EEG data). Hurst exponents were determined as the slope parameter of the linear fit to the detrended fluctuation function. DFA was applied to random walks that were constructed by partitioning the discrete variables (Potts model states, EEG microstate classes) into two subsets, and substituting these categorical discrete variables with the values ± 1, respectively (Van de Ville et al, 2010). There are three different $(2, 2)$-partitions for datasets with four states (Potts model for $Q = 4$, EEG microstates) and ten $(2, 3)$-partitions for five states (Potts model for $Q = 5$). For each time series, Hurst exponents were computed for each partition and the average across all partitions was used for statistical analysis.

Code availability

Sample Potts model data and analysis scripts to reproduce simplified versions of Figures 3-5 are available online (github repository).

Results

Potts model

Entropy rate and excess entropy

Results for the Potts model are shown in Fig. 3-Fig. 5. In Fig. 3, entropy rate (black) and excess entropy (blue) are plotted against relative temperature (T/T_c) for $Q = 4$ states (A) and $Q = 5$ states (B). The critical temperature is defined by $T/T_c = 1$ on the x-axis. Four- and five-state models ($Q = 4, 5$) gave very similar results.

With increasing temperature the entropy rate rises sigmoidally and the steepest slope occurs at the critical point. At low temperatures ($T/T_c = 0.2$), the time courses at most lattice sites are constant, or have very few state changes. As there is no randomness in the data, there is no prediction uncertainty and the entropy rate is close to zero. The maximum entropy rate for a Q-state process is $\log_2(Q)$ bits/sample, or 2.0 bits/sample for $Q = 4$ and 2.32 bits/sample for $Q = 5$. Both Potts models tend towards their respective maximum entropy rate value, indicating that the high temperature dynamics are largely random.

Excess entropy peaks at the critical point and decays to lower, but non-zero values, away from the critical temperature ($T < T_c$ and $T > T_c$). Although asymmetric around the critical temperature, the shape of the excess entropy curve reflects the concept of statistical complexity whereas entropy rate reflects the Kolmogorov complexity concept.

Above the critical temperature, the variability of both metrics was so small that the error bars (standard deviation) are not visible.

For both metrics, the Markov surrogate datasets (grey triangles) show only minor deviations from the actual model data results. Error bars for surrogate data are equally small but were omitted from the graph to improve visibility.

Lempel-Ziv complexity (LZC)

LZC analysis of Potts model data is shown in Fig. 4. LZC values, when scaled to bits/sample, are numerically very close to the entropy rate estimates shown in Fig. 3. Again, the steepest increase is found at the critical temperature and first-order Markov surrogates (grey triangles) are almost indistinguishable from model data for both $Q = 4, 5$. Minor deviations are observed around the critical temperature ($T/T_c = 1.0, 1.1, 1.2$) where surrogate data LZC is slightly larger than model data complexity. Error bars (standard deviation) are hardly visible due to an extremely low LZC variability. Variability across the 1250 surrogate time series was also very low, and error bars were omitted.

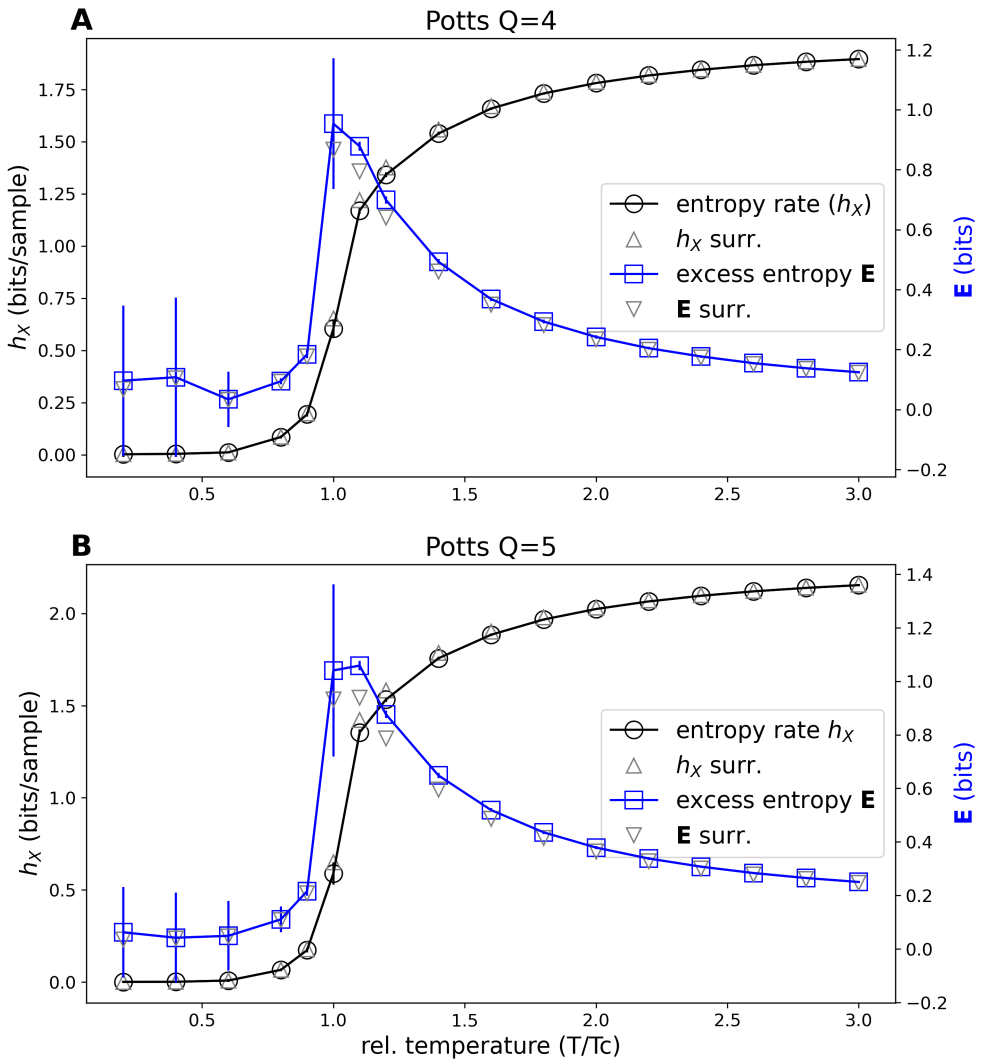


Fig. 3 Entropy-based complexity metrics for the Potts model with $Q = 4$ (A) and $Q = 5$ (B) states. The same pattern is observed for both models. The entropy rate (h_X, black lines and circles) is a strictly increasing function over the model control parameter (temperature). The steepest increase is observed around the critical temperature (relative temperature $T/T_c = 1.0$). The excess entropy (**E**, blue lines and squares) peaks close to the critical point and decreases away from the point of the phase transition ($T < T_c$ and $T > T_c$). The corresponding Markov surrogates (grey triangles) lie very close to the values computed from the actual data for all cases, $Q = 4, 5$, for h_X and **E**. Error bars represent standard deviations. Error bars were omitted for surrogate statistics to simplify the visual presentation.

Hurst exponents

Hurst exponents computed by DFA are shown in Fig. 5. For both Potts models, Hurst exponents peak at the critical temperature and decay towards $H = 0.5$ at high

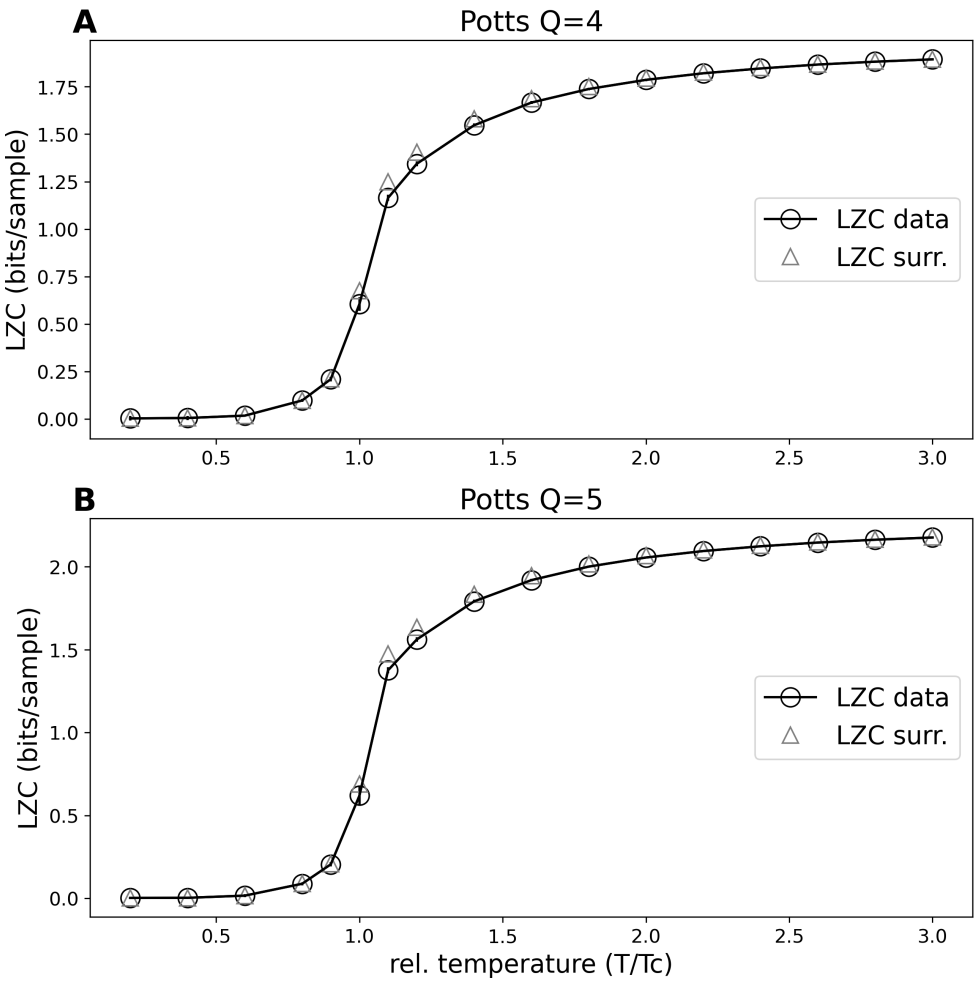


Fig. 4 Potts model Lempel-Ziv complexity (LZC) for $Q = 4$ (A) and $Q = 5$ (B). The same shape is observed for both models. LZC shows a monotonic increase with increasing system temperature, measuring the system's randomness, and the maximum slope is found at the critical point ($T/T_c = 1.0$). The numerical values are practically identical to the entropy rate values in Fig. 3. Markovian surrogate data produce almost identical LZC values (grey triangles).

temperatures, the theoretical value for random processes. Hurst exponent variability in the low temperature range was large and only the upper half of the error bars is shown. At low temperatures, many lattice sites experienced very few state changes which appear as large jumps in the random walk embedding. These jumps appear as significant fluctuations at long time scales, leading to steep fluctuation functions and large Hurst exponents.

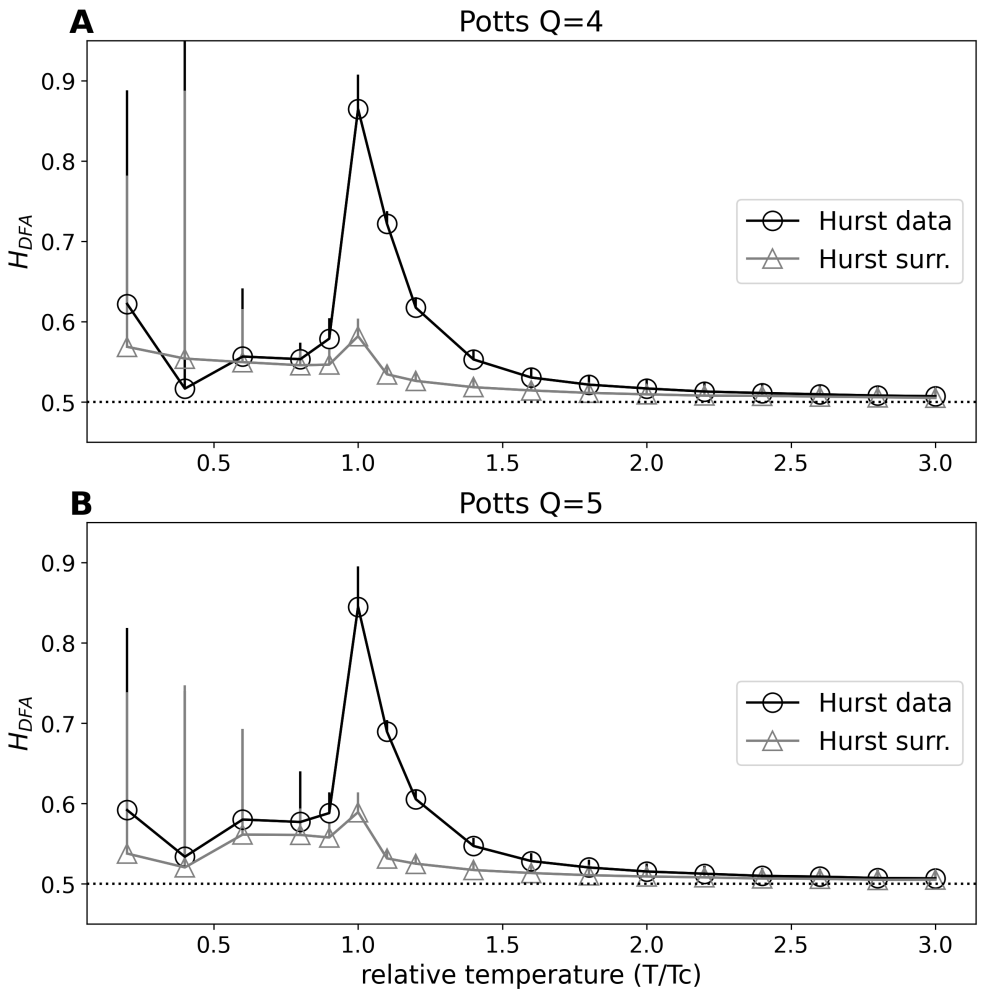


Fig. 5 Hurst exponents (H) measured by detrended fluctuation analysis (DFA) for the Potts model with $Q = 4$ (A) and $Q = 5$ (B). Both models ($Q = 4, 5$) produce highly similar curves. Hurst exponents computed on model data peak at the critical temperature ($T/T_c = 1.0$), representing long-range correlations as a different complexity marker compared to entropy and LZC. This behaviour is similar to the excess entropy metric shown in Fig. 3. Surrogate data (grey triangles) are clearly distinguished from actual data by smaller Hurst exponents, although a peak around the critical point is also observed.

EEG microstate sequences in wake and sleep

Entropy rate and excess entropy

Fig. 6 summarizes the results of entropy rate (A) and excess entropy (B) statistics for EEG microstate sequences in wake and sleep. While wakefulness and N1 sleep have a similar distribution of entropy rate (A) and excess entropy (B) values, entropy rate

values decrease with deepening sleep (N2, N3) and excess entropy increases. Results for EEG data (dark grey) and first-order surrogates (light grey) are similar. Two-way ANOVA analysis did not reveal any significant differences between original and surrogate data for entropy rate ($p = 0.828$) and excess entropy (p=0.696). One-way ANOVA and post-hoc analysis over the original EEG dataset revealed significant differences ($p_{FWE} < 0.05$) between all vigilance states except W vs. N1. This significance pattern was found for both metrics, entropy rate and excess entropy.

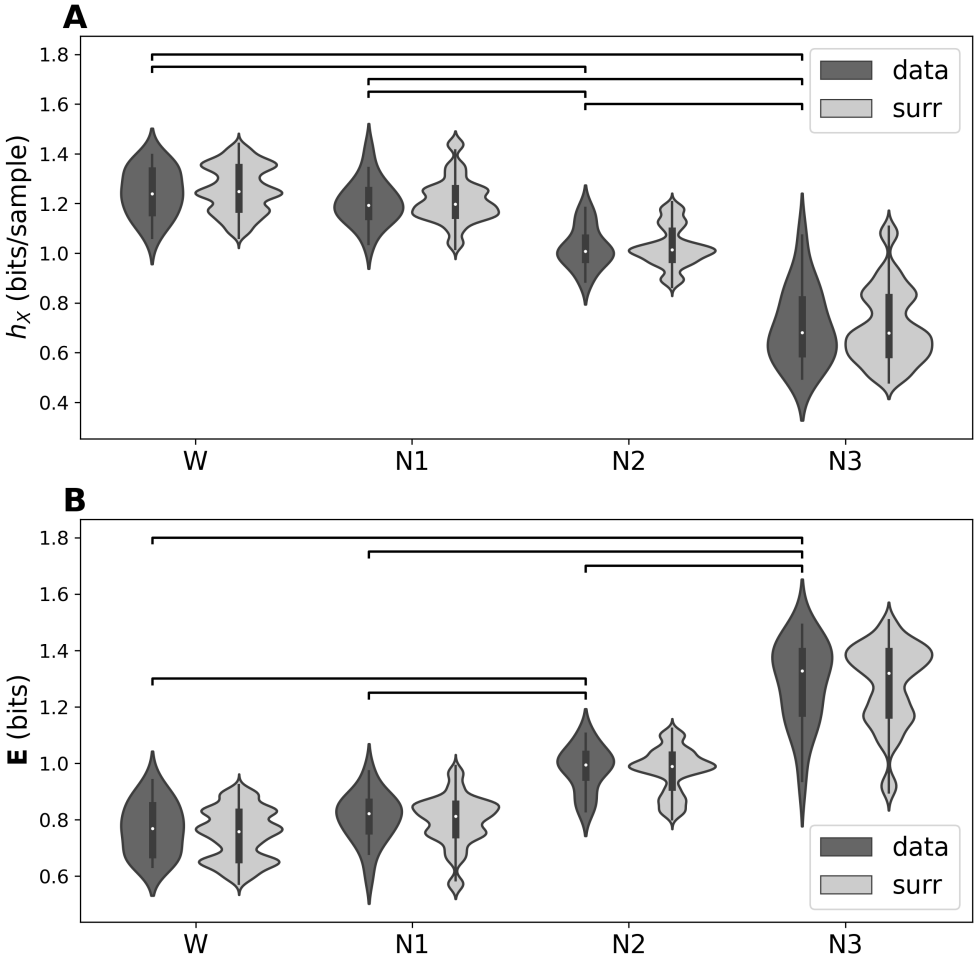


Fig. 6 Entropy rate (A) and excess entropy (B) metrics for EEG microstate sequences in wake and NREM sleep. Towards deeper sleep stages, entropy rate (h_X) decreases (A) and excess entropy (**E**) increases (B), suggesting less randomness and stronger pattern formation in sleep-related time series. Two-way ANOVA did not show differences between Markov surrogates ('surr', light grey) and the metrics computed on EEG microstate data ('data', dark grey), for entropy rate and excess entropy. One-way ANOVA of data-based metrics showed significant differences for entropy rate and excess entropy statistics ($p < 0.05$) and significant pairwise differences are indicated by brackets.

Lempel-Ziv complexity (LZC)

Lempel-Ziv complexity analysis of EEG microstate sequences in wake and sleep is shown in Fig. 7. The numerical LZC values were rescaled to bits/sample and were numerically very close to the entropy estimates in Fig. 6. LZC values for first-order surrogates (light grey) were statistically not different from original data LZC values (two-way ANOVA, p=0.881). When original data LZC values were analyzed separately, post-hoc comparisons showed significant differences ($p_{FWE} < 0.05$) between all vigilance states except for the W-N1 comparison.

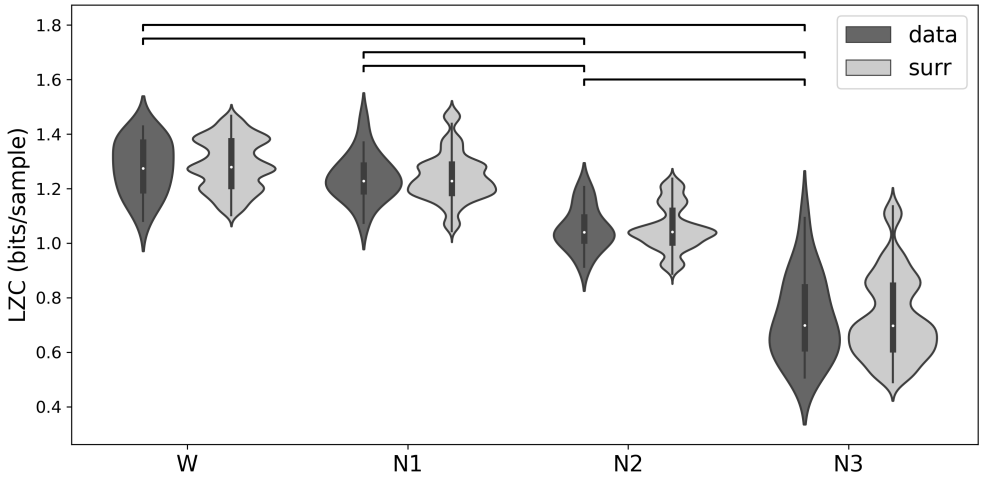


Fig. 7 Lempel-Ziv complexity for EEG microstate sequences in wake and NREM sleep. LZC decreases towards deeper sleep stages. The scaled LZC values from the original LZ-76 algorithm closely reproduce the entropy rate values in Fig. 6. Markovian surrogate data ('surr', light grey) statistically were not different from EEG microstate sequence LZC values ('data', dark grey), as shown by two-way ANOVA. One-way ANOVA of EEG data LZC values showed significant differences (one-way ANOVA, $p < 0.05$). Pairwise differences are shown by brackets.

Hurst exponents

Hurst exponent analysis of EEG microstate sequences in wake and sleep is illustrated in Fig. 8. The difference between EEG data and surrogate data Hurst exponents was significant (two-way ANOVA, $p = 0.001$). One-way ANOVA of Hurst exponents for original EEG data was significant ($p < 0.05$), and pairwise post-hoc analysis revealed significant differences between all vigilance states except for the W-N2 contrast. The mean Hurst exponent in wakefulness was slightly larger than in N1 (difference: -0.034), while N2 and N3 showed larger Hurst exponents than wakefulness although the differences between the means were small (N2-W: 0.014, N3-W: 0.062).

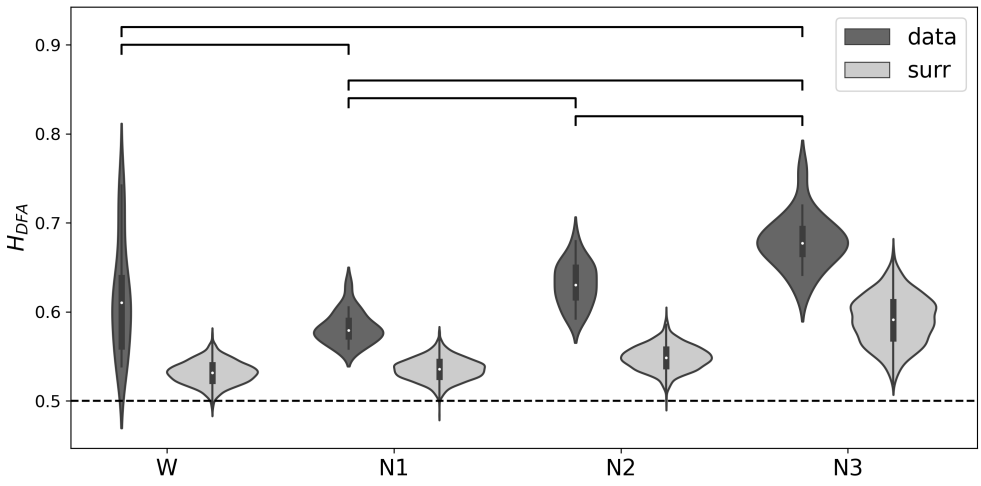


Fig. 8 Hurst exponents for EEG microstate sequences in wake and NREM sleep. Hurst exponents in wakefulness (W) show the largest variability and increase towards deeper sleep stages. Surrogate data ('surr', light grey) had lower Hurst exponents in general, and the difference between the factors 'data' and 'surrogates' was significant (two-way ANOVA, p=0.001). One-way ANOVA of EEG data Hurst exponents was significant ($p < 0.05$). Pairwise differences ($p_{FWE} < 0.05$) are illustrated by brackets.

Discussion

The main results of this report can be summarized as follows:

- Kolmogorov complexity and statistical complexity are two fundamentally different concepts of complexity and the existing EEG microstate analyses have focused on the former concept (randomness).
- We evaluated excess entropy for microstate sequence analysis and found that it (i) correctly identifies the critical point in the Potts model and (ii) increases with deepening NREM sleep stage in EEG microstate sequences.
- Rescaled Lempel-Ziv complexity and entropy rate are equivalent metrics of microstate complexity if the LZ-76 implementation is used. They represent the Kolmogorov complexity concept.
- Entropy-related measures including LZC are determined by first-order time series properties whereas Hurst exponents integrate information from longer time scales. Limitations of Hurst exponent analysis were observed for the low-temperature Potts model.

The Potts model and EEG microstate sequences

We decided to evaluate the selected complexity metrics on Potts model data before studying EEG data. The Potts model is a widely studied system in statistical physics with a discrete state space and a well-defined phase transition (Wu, 1982). The rationale for this approach was that it allowed us to gauge the behaviour of complexity

metrics at defined system states whereas the ground truth complexity of a brain state measured in an experiment is unknown. We used the Potts model variants with $Q = 4$ and $Q = 5$ discrete states, respectively, for two reasons. First, most EEG microstate analyses use either four or five microstate classes (Michel and Koenig, 2018). The second reason was that the 2D Potts model for $Q = 4$ undergoes a second-order phase transition and a first-order phase transition for $Q = 5$. Although there is evidence that resting-state brain activity shares phase-transition like features with physical models (Chialvo, 2010; Tagliazucchi et al, 2012), the type of phase transition occurring in the brain is unknown in general. Overall, all tested metrics showed the same characteristics for $Q = 4, 5$, except for the anticipated differences in absolute entropy values which depend on Q. We therefore conclude that the tested metrics do not depend on the type of phase transition.

Entropy rate and excess entropy

We have found entropy rate and excess entropy to reflect the concepts of Kolmogorov complexity and statistical complexity, respectively. Their difference can be expressed in terms of forecasting (Grassberger, 1986). Kolmogorov complexity describes how hard it is to predict a signal correctly, and statistical complexity measures how hard it is to find the best predictor. The difference between these two approaches becomes evident when applied to purely random patterns. They are impossible to predict precisely (high entropy rate and LZC), but the best predictor can be found easily. A microstate sequence without any temporal order, for example, is hard to predict but the best estimator is a simple bet on the relative probabilities of occurrence of each microstate, e.g. 1/4 each if uniformly distributed. Statistically complex patterns, however, can be easier to predict, but the best predictor can be impossible to find as it might need to capture the full non-linear dynamics of the underlying system.

One of the aims of this article was to explore excess entropy as a potentially useful addition to EEG microstate analysis. This raises the question whether excess entropy can contribute anything to the information already gained from other metrics. The answer is that excess entropy reflects the concept of statistical complexity which is not captured by either entropy rate or Lempel-Ziv complexity, and that its value cannot be predicted from these other metrics. The latter point is illustrated by the Potts model results which show that excess entropy and entropy rate can covary either positively or negatively. For sub-critical temperatures ($T/T_c < 1$ in Fig. 3), both metrics increase with temperature. At low temperatures, future states are not difficult to predict as there are few state switches and conditioning on the past hardly improves the prediction. Approaching the critical temperature, future states become increasingly difficult to predict (higher entropy rate) but at the same time knowledge of the past significantly improves the prediction (higher excess entropy). Improved predictability is explained by expanding autocorrelations near the phase transition which can be exploited to make predictions less uncertain. In information-theoretic terms, we find more shared information between the past and the future states of the sequence near the critical point. Above the critical temperature, randomness (entropy rate) increases further but dependencies on past states fade as sequences approach the uncorrelated noise level.

Sample entropy, a measure related to entropy rate, was evaluated in (Murphy et al, 2020) where an almost random pattern of microstates in psychosis patients was observed. It must be noted, however, that a key difference between microstate studies is the type of input signal, which can either be the full sequence of microstate labels at each time step (von Wegner et al, 2018a; Jia et al, 2021; Wiemers et al, 2022) or the same sequence after removal of duplicate labels (Murphy et al, 2020). The sequence without duplicates, where only state changes are retained, has received different names. The common name in Markov process theory is the jump process (Gillespie, 1992), it was called transitioning sequence in (Tait et al, 2020) and no-permanence sequence in (Artoni et al, 2022). The entropy rate of the jump chain is higher than that of the full sequence as duplicates reduce the apparent randomness of the signal. For microstate research there is no strict rule which signal should be preferred. For a pure Markov process, the choice is hardly relevant as the relation between the full and the jump processes is completely defined by the transition rates (Gillespie, 1992). This is not the case for most EEG microstate sequences which are non-Markovian (von Wegner et al, 2017). The distribution of the discarded duplicate segments in that case is unknown and potentially conveys information about the experimental condition. We therefore prefer to work with full sequences.

Lempel-Ziv Complexity

A collection of data compression algorithms developed in the 1970s are associated with the names of Lempel and Ziv, notably the LZ-76, LZ-77, and LZ-78 algorithms (Lempel and Ziv, 1976, 1977; Ziv and Lempel, 1978). Despite their similar names, only the LZ-76 algorithm has a close relationship to theoretical quantities like the entropy rate, whereas later versions were optimized for practical data compression tasks. This has been shown for neural spike train data where the earlier LZ-76 algorithm proved to be numerically equivalent to the entropy rate of binarized spike trains, whereas LZ-78 contained a marked bias relative to the true entropy rate (Amigó et al, 2004). In EEG microstate research, Lempel-Ziv complexity (LZC) was first used by Tait et al. (Tait et al, 2020), and subsequently by (Artoni et al, 2022). The former article uses the authors' LZ-76 implementation (plus-microstate toolbox) whereas the latter used the dictionary-based LZMA2 algorithm from the popular 7zip software (7zip). In analogy to (Amigó et al, 2004), we found that normalized LZC output from the LZ-76 algorithm provided excellent numerical agreement with the entropy rates computed from joint probability distributions as evident from comparison between Fig. 3 and Fig. 4. Entropy rate and LZC have so far been discussed separately in the microstate literature (von Wegner et al, 2018a; Jia et al, 2021; Tait et al, 2020; Artoni et al, 2022; Wiemers et al, 2022). The equivalence of both metrics means that LZC values as those given in (Tait et al, 2020) and entropy values we published in different contexts (von Wegner and Laufs, 2018; Jia et al, 2021; Wiemers et al, 2022) can be re-scaled and compared directly. Unfortunately, this is not possible for LZC values computed by more recent versions of the LZ algorithm, for example the complexity values given in (Artoni et al, 2022). Complexity metrics computed from the LZ-77 and LZ-78 algorithms are likely to yield similarly shaped curves to those in Fig. 3 and Fig. 4, but

the LZ-76 implementation has the advantage of being a direct numerical estimator of the theoretically meaningful entropy rate.

Conversely, the LZ-76 algorithm is fast and computationally less expensive than joint entropy estimations and is therefore an appealing practical alternative to computing entropy rates. This has potential applications for statistical comparisons of EEG microstate sequences with non-trivial surrogate data. To compare the Kolmogorov complexity of microstate sequences and Markov surrogates, the surrogate entropy rate can be calculated directly from the transition matrix, avoiding the cost of synthesizing a large number of surrogate time series, and the data entropy rate could be obtained via LZC.

The joint entropy approach, on the other hand, still has the advantages that (i) it returns excess entropy 'for free' from the same linear fit procedure that is used to obtain the entropy rate, and (ii) the joint entropy values for microstate words of a given length might be of interest for certain research questions. A further argument in favor of the LZ-76 algorithm is that there are no free parameters that need to be set by the researcher and that will affect the numerical output of the algorithm. The free parameters of the LZMA2 algorithm (compression level, dictionary size, number of fast bytes, filter and match-finder options, (Artoni et al, 2022)) are useful to achieve a maximum compression level for different data types but they have no evident theoretical meaning and preclude numerical agreement with the actual entropy rate.

In summary, all LZ implementations could be used to estimate the Kolmogorov complexity of microstate sequences but the LZ-76 implementation has the advantage that its output can be interpreted as an entropy rate.

Hurst exponents

Fractal dimension and Hurst exponent analysis are based on a different theoretical framework than entropy-based methods, and they can be found as complexity measures in functional MRI studies (Bullmore et al, 2009; Tagliazucchi et al, 2013; Dong et al, 2018), EEG studies (Linkenkaer-Hansen et al, 2001; Raghavendra et al, 2009; Sabeti et al, 2009; Holloway et al, 2014), but also outside brain research (Raubitzek and Neubauer, 2021).

Applied to Potts model data, Hurst exponents represent statistical complexity in a similar way to excess entropy. In contrast to the entropy-related metrics including LZC, Hurst exponents for Markov surrogates had markedly lower Hurst exponents. This could be observed for Potts model data (Fig. 5) as well as for EEG microstate sequences (Fig. 8). This demonstrates that the inclusion of larger time scales up to 10 seconds in the DFA parameters identified time series properties that were not visible to the entropy-based methods. This observation may appear obvious because entropy rate/excess entropy calculations were based on a finite history length of only $k = 6$ samples. Lempel-Ziv analysis, however, does not contain an explicit time window and progressively scans the whole time series and keeps track of previously seen patterns. Yet, LZC values were identical to entropy rate values which shows that there was no additional information on larger time scales through the Lempel-Ziv lens. In contrast to Lempel-Ziv analysis, DFA does not search for actual repetitions of exact patterns but accepts anything that contributes variance at a given time scale. Although DFA

quantifies self-similarity in signals that exhibit this property, it is not a generic detector of self-similarity in the strict sense. We can therefore conclude from Fig. 5 and Fig. 8 that Potts model data and EEG microstate sequences in deeper sleep stages display slow fluctuations that are not explained by exact repetitions of signal segments.

Potts model data analysis also demonstrated an important pitfall of DFA and Hurst exponent analysis. The large variability of Hurst exponents observed for the cold Potts model in Fig. 5 is caused by time series with a very small number of state changes. These cause isolated step-like jumps when the signal is transformed into a random walk according to the technique proposed for microstate sequences (Van de Ville et al, 2010). Jumps in turn lead to large fluctuations at long time time scales and empirically result in large Hurst exponents $H > 1.5$ although the signal is constant most of the time. When zero state changes occur, however, the expected value of $H = 0.5$ was obtained. This leads to the undesired result that near-constant time series that differ only in a small fraction of values result in massively different Hurst exponents. Again, this is related to the DFA algorithm being a variance analyzer. Non-stationarities such as steps in the signal add variance over a range of time scales that could be misinterpreted as a monofractal signature from the value of the Hurst exponent alone. When a monofractal time series like fractional Gaussian noise is analyzed with DFA, its fractal structure will be encoded by the Hurst exponent, but inverting this inference is not a valid procedure. When an arbitrary time series is analyzed with DFA, a Hurst exponent $H > 0.5$ does not necessarily indicate fractality or self-similarity (von Wegner et al, 2018b). It should also be noted that we have not encountered near-constant time courses over hundreds or thousands of samples for EEG microstate sequences, and we believe that the problem discussed for the cold Potts model is not an issue for EEG microstate analysis if the sequence length is at least a few hundred samples.

Finally, the transformation of discrete sequences into binary signals to construct a random walk that can be analyzed by DFA is still open to the same issues we discussed in (von Wegner et al, 2016). We followed the technique proposed in (Van de Ville et al, 2010), which is to assign the (categorical) microstate classes to a metric variable (± 1), and to average across all possible partitions of microstate classes. The pooling of different microstates, or cortical activity patterns, into a single class raises the still unanswered question how this class should be interpreted neurobiologically. Moreover, when five (Shi et al, 2020; Artoni et al, 2022) or seven (Jia et al, 2021) microstate classes are used, partitions are necessarily unbalanced, e.g. $((A, C), (B, D, E))$. Despite these reservations, our practical use of this approach has produced results concordant with other metrics and behavioural variables (Jia et al, 2021).

We conclude that the Hurst exponent of a microstate sequence can (i) be interpreted as a marker of statistical rather than algorithmic complexity, and (ii) provides a perspective on short and long time scales whereas the entropy-related measures we evaluated reflect short-range dependencies.

Microstate sequence complexity in wake and sleep

In this study, we have applied excess entropy, LZC, and DFA as novel methods to characterize the complexity of microstate sequences from wakefulness and NREM sleep. Entropy rate analysis was presented in (Wiemers et al, 2022) but is shown again here

with additional Markov surrogate comparisons. Our main findings are that deepening sleep stages were accompanied by decreasing entropy rate and LZC, increased excess entropy, and increasing Hurst exponents.

The reduced entropy rate and LZC values are probably related to the progressive slowing of EEG rhythms in sleep. Longer microstate duration (Wiemers et al, 2022) improves the predictability of microstate sequences in sleep stages N2 and N3 and reduce their apparent randomness. This result is similar to the findings of Tait et al. who found EEG slowing and reduced LZC values in Alzheimer disease patients although they corrected their microstate analysis for background EEG slowing effects (Tait et al, 2020; Dauwels et al, 2011).

From the discussion of the Potts model it is clear that the observation of entropy rate changes cannot predict changes in excess entropy. We found increased excess entropy values in N2 and N3, which corresponds to the Potts model situation above the critical temperature. Within this analogy, the transition to deeper sleep stages corresponds to a shift from higher temperatures towards the critical point. This is intuitively correct as deep sleep is accompanied by EEG signals with longer, delta rhythm-related autocorrelations which translate into increased shared information between past and future microstates. This explanation is further substantiated by our recent finding that microstate oscillations in the delta frequency range occur in N3 (Wiemers et al, 2022), and by the larger Hurst exponents in N2 and N3 (Fig. 8). The straightforward conclusion from this analogy would be to classify deeper sleep stages as being closer to a critical system state. This, however, at least partially contradicts earlier studies using EEG and other imaging modalities.

Weiss et al. found larger Hurst exponents in deep sleep stages using the DFA-related R/S statistic on raw EEG data from humans (Weiss et al, 2009). In rats, local field potential measurements revealed a similar trend, i.e. a lower complexity (LZC) in NREM vs. wakefulness (Abásolo et al, 2014). Another measure called Omega complexity, based on the eigenvalue spectrum of the spatial principal component analysis of an EEG data set, is known to decrease in deeper NREM sleep stages as well (Wackermann, 1996). Not all analyses point in the same direction however. Kantelhardt et al. performed an extensive analysis of Hurst exponents across sleep stages and frequency bands and found different trends for different frequency bands. In the alpha band, the main contributor to microstate topographies (Milz et al, 2017), the authors found lower Hurst exponents in deeper sleep, contradicting our results for microstate sequences. The opposite trend, however, was reported for other frequency bands (Kantelhardt et al, 2015). Different outcomes might also be due to the pre-processing of the input signals. Frequency band analyses often use the power (envelope) of narrow frequency band oscillations and not the oscillatory signal itself (Linkenkaer-Hansen et al, 2001; Kantelhardt et al, 2015), whereas microstate analysis starts with broad-band EEG oscillations and does not use the envelope signal. When a different recording modality is used, the results can change once again. In the past we have observed smaller Hurst exponents during sleep in regional BOLD (blood oxygen level dependent) signals from functional MRI data, and interpreted those as a departure from criticality, concluding that wakefulness was the state closest to criticality (Tagliazucchi et al, 2013). The methodological differences and the nature of the physiological signal used

might explain why electrophysiological and imaging data are unable to give a unique answer to the question how close the brain is to a critical state in a defined condition.

Another difference between the Potts model and EEG data is that the Potts model is controlled by a single parameter (temperature) that, when varied continuously, controls the extent of autocorrelations in the data. Sleep stage transitions, on the other hand, involve a switching between different system states in which different frequency generating circuits are active, namely alpha generators in wakefulness and delta/slow wave generators in N3. Furthermore, sleep contains isolated events never observed in wakefulness (vertex sharp waves, sleep spindles, K-complexes) (Adamantidis et al, 2019). From electrophysiological studies we know that different neuron populations partake in these patterns and that their voltage responses are fundamentally different, e.g. spiking vs. bursting (Llinás and Steriade, 2006). Similar heterogeneity is not modelled by the Potts system and demonstrates limitations of statistical physics models of brain activity. A shift from alpha to delta frequency band activity, with corresponding repercussions on microstate patterns as observed in (Wiemers et al, 2022) alone can probably explain a reduced entropy rate, increased excess entropy and larger Hurst exponents due to larger variance contributions at long time scales. The interpretations with regard to Kolmogorov and statistical complexity are unequivocal, yet, in the absence of an experimentally controllable parameter we would hesitate to strictly interpret this as the same system moving closer to a phase transition. This might be possible in modeling studies though, which indicate that phase transition-like mechanisms might indeed play a role in wake-sleep and anesthesia-induced transitions (Steyn-Ross et al, 2004, 2005). Further discussions about the complexity of brain activity during sleep can be found in (Olbrich et al, 2011) and references therein.

Conclusion

The concepts of Kolmogorov and statistical complexity are useful to structure future discussions about EEG microstate sequence complexity. We hope that the results of this study, namely the evaluation of excess entropy, the demonstrated equivalence between entropy rate and LZC, and the systematic comparison with Hurst exponent analysis will be of practical value to microstate researchers.

Acknowledgments. The authors would like to thank Luke Tait for providing generous insight into the analyses used in (Tait et al, 2020).

Declarations

The authors declare that they have no conflict of interest. This work was funded by the German Research Foundation (DFG), project number 440536202.

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